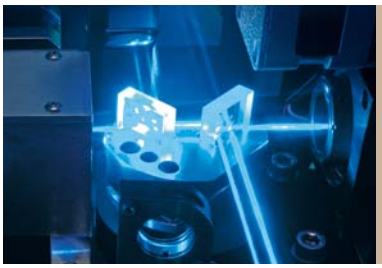

Q-switched DPSS Lasers

EO-Q-YLF

Nd:YLF DIODE-PUMPED Q-SWITCHED LASERS

Quantas EO-Q-YLF is compact, air-cooled, Q-switched laser designed for wide range of applications that require low pulse repetition rate and high peak power pulses (EO-Q1D-YLF model produces ~6 MW peak power).

- **Up to 40 mJ pulse energy at 1053 nm**
- **Air cooled**
- **Variable pulse repetition rate**
- **Build-in sync pulse generator for triggering of user equipment**
- **Remote control via build-in Ethernet interface**
- **Optional 2nd, 3rd, 4th or 5th harmonic generators**
- **Optional attenuator for 1053 nm, 527 nm and 351 nm wavelengths**
- **Optional fiber coupled output**
- **OEM version available**

APPLICATIONS

- **Light Induced Breakdown Spectroscopy (LIBS)**
- **Time-of-Flight Spectroscopy (TOFS)**
- **Light Induced Fluorescence (LIF) Spectroscopy**
- **Flash photolysis**
- **Matrix Assisted Laser Desorption/Ionization (MALDI)**
- **Pulsed light deposition (PLD)**
- **Remote sensing**
- **Laser ablation**

SPECIFICATIONS [1]

Model	Quantas EO-Q1A-YLF	Quantas EO-Q1B-YLF	Quantas EO-Q1C-YLF	Quantas EO-Q1D-YLF
Wavelength	1053 nm [2]			
Pulse energy				
Fundamental	5 mJ	10 mJ	20 mJ	40 mJ
2nd harmonic (527 nm)	2.3 mJ	5 mJ	10 mJ	20 mJ
3rd harmonic (351 nm)	1.2 mJ	3 mJ	6 mJ	12 mJ
4th harmonic (263 nm)	0.6 mJ	1.5 mJ	3 mJ	6 mJ
5th harmonic (211 nm)	0.15 mJ	0.5 mJ	1 mJ	2 mJ
Pulse repetition rate	Single shot – 10 Hz [3]			
Pulse duration	< 8 ns [4]			
Pulse-to-pulse energy stability	< 1 % rms [5]			
Power drift	± 3% [6]			
Beam profile	Nearly TEM_{00}		Bell-shaped, > 75 % fit to Gaussian	
Beam divergence [7]	< 3 mrad		< 1.5 mrad	
Beam diameter, typical [8]	1.2 mm	1.5 mm	2 mm	2.5 mm
Polarization	Linear, horizontal			
Optical jitter	N/A	< 1 ns rms [9]		
PHYSICAL CHARACTERISTICS & OPERATING REQUIREMENTS				
Laser head (W × L × H)	113 × 230 × 112 mm			
Harmonic generator (W × L × H)	113 × 242 × 112 mm			
Controller unit (W × L × H)	85 × 165 × 50 mm			
Power adapter, typical (W × L × H)	50 × 125 × 31 mm			
Operation environment	15–30 °C, 10–80 % humidity – non-condensing 90–240 V AC, 30 W, 47–63 Hz [10]			

1) *The parameters marked typical are not specifications. They are indications of typical performance and might vary unit-to-unit. Unless stated otherwise all specifications are measured at 1053 nm at 10 Hz pulse repetition rate.*
2) *1064 nm output wavelength models are available. Check EO-Q-YAG series.*
3) *Factory-set pulse repetition rate is fixed at 10 Hz. Single shot or variable pulse repetition rate is possible when laser is externally triggered. Higher repetition rates are available, please inquire for details.*
4) *FWHM at 1053 nm. Pulse durations as short as 3 ns are available by request. Please inquire for detailed specifications.*
5) *Averaged from 500 pulses.*
6) *Over 8 hour period after 20 minutes of warm-up when ambient temperature variation is less than ±2 °C.*
7) *Full angle measured at the $1/e^2$ level.*
8) *Beam diameter is measured 20 cm from laser output at the $1/e^2$ level.*
9) *In respect to Q-switch triggering edge of the pulse.*
10) *Laser can be powered from appropriate 12 VDC power source. Please inquire for details.*